

Практикум B5

85 marks, 37 blocks, nine techniques.

Every question in this topic is one of two sentences. The unknown is in the exponent — and the only way down is a logarithm. The unknown is under a logarithm — and the only way up is a power. Everything else is arithmetic.

О практикуме

Четвёртый практикум серии на английском, устроенный как B2-B4: ноутбук, карточка приёмов и сообщения проверок английские, документация репозитория остаётся русской.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-b5-exponential-log-models

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Practicum B5: exponential and logarithmic models

85 marks, 37 blocks, nine techniques. Every question in this topic is one of two sentences. *The unknown is in the exponent* — and the only way down is a logarithm. *The unknown is under a logarithm* — and the only way up is a power. Everything else is arithmetic.

Material. The whole of `functions.logarithmic_functions`, `functions.exponential_models`, `functions.mathematical_models` and `number_algebra.exponential_models` from the AA HL archive, sessions May 2021 — November 2025. Four topics, because no one of them reaches the lower bound of 60 marks on its own — the largest is 35 — and because the archive does not separate them either: November 2025 Paper 3 fits a logistic curve, takes logarithms of it, and then solves the

differential equation behind it, in one question.

This practicum is in English, like B2, B3 and B4. The checks speak whichever language the notebook asks them to, and this one asks for English in the setup cell.

The two sentences.

$3000 = 1000e^{8k}$ — *the unknown is in the exponent.* Take logarithms of both sides and it walks down: $k = \frac{\ln 3}{8}$.

$\log_3(x + 6) + \log_3 x = 3$ — *the unknown is under a logarithm.* Make the bases agree, collapse each side to one logarithm, and then remove them: $x = 3$.

Everything in the topic is one of these with more furniture around it. A population model is the first sentence with two data points instead of one. Compound interest is the first sentence with the base written as $1 + r$. Decibels

are the second sentence with a factor of 10 in front.

Where the calculator sits, and here it is not an illusion. 54% of the marks carry one, and unlike B2 and B3 the split is real: Paper 1 asks for the laws and the exact form, Paper 2 and Paper 3 ask you to fit a model to data and predict. The same technique — take logs of both sides — ends at $\frac{\ln 2}{5730}$ on one paper and at 2380 on the other.

How the checks work here. Two of them are new.

- `verify_model` takes the model you fitted and puts the data back into it. No stored constants: your A and k are right when $A(0)$ and $A(8)$ come out to what the question said they were. This is where the marks in the topic actually go — *"recognizing that $t = 8$ on 1 January 2022"* is one of the seven marks in November 2022, and getting t wrong is the standard way to lose the question with perfect algebra.

- `verify_in_terms_of` is for *"find an expression for $\log_{10} 24$ in terms of p and q ".* It checks two things at once: that nothing but the given letters appears in your answer, and that substituting their true values gives the right number. Without the first check, $\log_{10} 24$ rewritten as itself would pass.

How to work

1. Read the map of techniques first. It is arranged by **which way the unknown has to move**.
2. Work **on paper** before typing anything. On Paper 1 questions there is no calculator at all.
3. Exact answers where the paper is Paper 1: `Rational(1, 9)`, `log(2)/5730`, `3*p + q`. Three significant figures where the paper is Paper 2 or 3.
4. A model is entered as an expression in `t`: `15000*exp(-0.0146*t), 200/(1 + 4*exp(-0.153*t))`.
5. A set of values is entered as a set or an inequality: `Interval(exp(-5), E)`, `(x >`

```
-1), Union(...).
```

6. The last three blocks are a Paper 1 / Paper 2 comparison, a recognition trainer, and one question on a timer.

Difficulty marks: ● the technique on its own · ● the technique in a wrapper · ● several techniques, or a whole exam question.

```

import sys
sys.path.append('.')          # from
practicum/functions to practicum/kit/
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *             # checks +
Interval, Union, solveset, Rational

language('en')                # this
notebook is in English, and so are the checks

a, b, n, p, q, r = symbols('a b n p q r')
S, T, v0 = symbols('S T v0', positive=True)
# S here is the sum from Task 12, and it
# shadows sympy's S. Where a check
# wants "all the reals", write Interval(-oo,
oo) rather than S.Reals.

print('ready; sympy', sp.__version__)
print('in terms of p and q: ', 3*p + q)
print('an exact log:          ', log(2)/5730)
print('a model in t:          ',
15000*exp(log(Rational(89, 100))/8*t))
print('a set of values:       ',
Interval(exp(-5), E))

```

Map of techniques

#	Technique	Trigger in the question	First move
1	The three log laws	several logs of one base; «in terms of p and q »	factorise the argument
2	Change of base	two different bases in one line	rewrite everything in the smallest base
3	Log equation	x under a logarithm	collapse each side to one log
4	Laws of exponents	powers of one base; a given relation like $e^T = 1 + v_0$	split the exponent
5	Take logs of both sides	the unknown is in the exponent	isolate the power first
6	Fit the model	a form with letters and two data points	decide what $t = 0$ means
7	Percentages as a power	«per annum», «compounded», «inflation»	turn the percentage into a multiplier

#	Technique	Trigger in the question	First move
8	The logistic model	a stated long-run limit L	put $t = 0$ and use $e^0 = 1$
9	The logarithm as a function	$\ln$ inside another function; a log scale	solve for $\ln x$, remove it last

The ladder goes by which way the unknown has to move.

Rungs 1-3 — upwards, out from under the logarithm. First the three laws, which are the whole of Paper 1 in this topic. Then change of base, which exists for one reason: until the bases agree, no law applies. Then the equation, which is the two of them plus one step — remove the logarithms — and one habit: check the arguments were positive.

Rungs 4-6 — downwards, out of the exponent. The laws of exponents first, because they are what makes the descent possible and because two questions in the archive are nothing else. Then the descent itself. Then the descent done twice, which is what fitting a model

is: one data point gives the multiplier, the second gives k .

Rungs 7-8 — the two models the archive actually asks about.

Percentage growth is $A(1+r)^t$, and the whole difficulty is translating English into r : *nominal 4% compounded quarterly* means $r = 0.01$ per quarter, and *inflation* means you divide by $1+j$ rather than subtracting j . The logistic model is $L/(1 + Ce^{-kt})$, and it is the same fit with a ceiling read straight off the page.

Rung 9 — the logarithm as a function, not as a tool.

$\ln$ inside a composition, $\ln$ defining a domain, $\ln$ as a measuring scale. This comes last not because it is hard but because it uses rung 1 in a place where you were not looking for it.

What saves the most time.

Before touching anything, write down what $t = 0$ means and what one unit of t is. Half the lost marks in this topic are there, and

they are lost before any mathematics happens.

Part I — upwards: out from under the logarithm

Theory 1. Three laws, and the one that does not exist

$\log_b a$ is the answer to one question: *to what power must b be raised to give a ?* Everything follows from that, including the three laws:

$$\log_b(MN) = \log_b M + \log_b N,$$

$$\log_b \frac{M}{N} = \log_b M - \log_b N,$$

$$\log_b M^r = r \log_b M.$$

The law that does not exist is $\log(M + N)$. There is no rule for it, and the examiner knows exactly how tempting it looks. Every appearance of a sum inside a logarithm in this archive is there to be *factorised* first: $\log_{10} 45 = \log_{10}(3^2 \cdot 5)$, not $\log_{10}(40 + 5)$.

The fourth fact, which is not a law but is worth as many marks. A number standing next to a logarithm has to become one before it can join in:

$$1 = \log_b b, \quad 3 = \log_2 8, \quad -1 = \log_{10} \frac{1}{10}.$$

Both log-equation questions in the archive hinge on this. November 2025 TZ1 has $\log_2[\dots] - 1 = \log_2[\dots]$, and the -1 is the question.

"In terms of p and q " is an instruction about the form of the answer. If a question gives $p = \log_{10} 2$ and $q = \log_{10} 3$ and asks for $\log_{10} \frac{9}{2}$, the answer is $2q - p$: factorise the argument, then read the laws right to left. A decimal, however accurate, is not an answer — and neither is the same logarithm rewritten as $\frac{\ln 4.5}{\ln 10}$, which has changed the base and answered nothing.

Monotonicity is a licence, and it costs a mark to skip. To go from $u \geq v > 0$ to $\log_2 u \geq \log_2 v$ you need the fact that $\log_2$ is increasing — the step is not algebra, it is a property of the function,

and it has to be named. The markscheme for May 2025 TZ1 Q8(a) awards **R1** for exactly that sentence.

Task 1 — the laws in both directions

(a) Let $\log_{10} 2 = p$ and $\log_{10} 3 = q$. Find an expression for $\log_{10} 24$ in terms of p and q .

(b) Show that $1 + \log_2 n \geq \log_2(n + 1)$ for $n \in \mathbb{Z}^+$.

For (b) the answer is printed in the question, so there is nothing to hide. What the cell checks is the step that earns the marks: write $1 + \log_2 n$ **as a single logarithm**.

May 2025 TZ3 Paper 1 Q1(a) — 3 marks.

May 2025 TZ1 Paper 1 Q8(a) — 2 marks.

```
qla = ...          # log_10 24 in terms of p
and q
qlb = ...          # 1 + log_2 n written as one
logarithm, e.g. log(<something>, 2)

verify_in_terms_of('la', qla, log(24, 10),
{p: log(2, 10), q: log(3, 10)})
verify_identity('lb', qlb, 1 + log(n, 2),
var=n)
```

Theory 2. Change of base, and why it is always available

$$\log_b a = \frac{\log_c a}{\log_c b} \quad \text{for any base } c > 0, c \neq 1.$$

Two logarithms of different bases cannot be added, subtracted or compared. Nothing from Theory 1 applies until they agree. So the first move in any question with two bases is to pick one and convert.

Which base to pick. Almost always the smallest one that the others are powers of — and if the question has already handed you a logarithm in some base, that base is the one it wants:

bases seen	base to use
5 and 25	5, since $25 = 5^2$
9 and 27	3, since $9 = 3^2$ and $27 = 3^3$
10 and something, with $\log_{10}$ already given	10, because the given value is what the answer must be built from

The shortcut worth memorising.

When the bases are powers of one number, the conversion is just a fraction:

$$\log_{b^m} a = \frac{1}{m} \log_b a.$$

That is why $\log_{16} a = \frac{1}{4} \log_2 a$ in one line — if $\log_2 a = 6$ then $\log_{16} a = \frac{3}{2}$, with no calculator anywhere — and why $5 \log_{32} y = \log_2 y$: the 5 in front and the 5 in $32 = 2^5$ cancel exactly.

The direction people get wrong.

$\log_b a = \frac{\log a}{\log b}$: the *argument* goes on top, the *base* goes underneath. Check it once on something you know — $\log_2 8 = \frac{\log 8}{\log 2} = \frac{3 \log 2}{\log 2} = 3$ — and you will not flip it again.

Why your calculator needs this too.

A GDC computes only $\ln$ and $\log_{10}$. Every other base it reaches by exactly this formula. It is not a trick for exams; it is the definition being used sideways.

Task 2 ● — one base at a time

(a) Let $\log_{10} 2 = p$ and $\log_{10} 3 = q$. Find an expression for $\log_3 8$ in terms of p and q .

It is given that $\log_{10} a = \frac{1}{3}$, where $a > 0$. Find the value of

(b) $\log_{10} \frac{1}{a}$; **(c)** $\log_{1000} a$.

May 2025 TZ3 Paper 1 Q1(b) — 2 marks.

May 2024 TZ1 Paper 1 Q2 — 5 marks. All Paper 1: the answers are exact.

```
q2a = ...          # log_3 8 in terms of p and q
q2b = ...          # log_10 (1/a)
q2c = ...          # log_1000 a

verify_in_terms_of('2a', q2a, log(8, 3), {p: log(2, 10), q: log(3, 10)})
verify_exact('2b', q2b, -Rational(1, 3))
verify_exact('2c', q2c, Rational(1, 9))
```

Theory 3. Solving when the unknown is under the logarithm

The recipe is fixed, and every step of it is a separate mark.

1. **Make the bases agree** (Theory 2).
2. **Collapse each side to a single logarithm** (Theory 1), turning any loose number into $\log_b b^{\text{that number}}$.
3. **Remove the logarithms.** From $\log_b A = \log_b B$ conclude $A = B$; from $\log_b A = c$ conclude $A = b^c$. This is legitimate because $\log_b$ is one-to-one — the same monotonicity that licensed the inequality in Task 1.
4. **Solve the algebraic equation** that is left.
5. **Check the arguments.** Every expression that stood inside a logarithm must be strictly positive at your root. This is not tidiness: taking logarithms is not reversible, and step 3 invents roots that the original equation never had.

Where the archive puts the difficulty.

Not in step 4. In both log-equation questions of the archive what is left after step 3 is one line of school algebra, and all the marks are in steps 1–3: making

the bases agree, turning a loose number into a logarithm, and collapsing each side to a single one. If step 4 is taking you more than a line, you have collapsed something wrongly.

The trap with a name. From $\log A - \log B = \log C$ people conclude $A - B = C$. The subtraction is *outside* the logarithms and the division is *inside*: $\frac{A}{B} = C$. Write the quotient down explicitly before removing anything.

Task 3 — two bases, one unknown

Solve the equation $3\log_8 10x - \log_4 x = 1$ for $x > 0$.

Enter the answer as a **list** of roots: `[Rational(1, 5)]`, `[4, 9]`. The check substitutes each of them into the equation itself and then scans the interval for the ones you did not write down.

November 2025 TZ3 Paper 1 Q4 — 5 marks, no calculator.

```
q3 = [...]          # all roots with x > 0

verify_roots('3', q3, 3*log(10*x, 8) - log(x,
4) - 1, (0.0001, 100))
```

Task 4 — the loose -1 , and the domain that removes a root

(a) Given that $x > 7$, show that

$$\frac{x}{x^2 - 8x + 7} \times \frac{x^2 - 1}{x + 1} \equiv \frac{x}{x - 7}.$$

(b) Hence, or otherwise, solve

$$\begin{aligned} \log_2[x(x^2 - 1)] - 1 \\ = \log_2[(x^2 - 8x + 7)(x + 1)]. \end{aligned}$$

Part (a) is a factorising exercise, and the cell asks for the factor that does the work: write $x^2 - 8x + 7$ as a product of two linear factors. Part (b) is the equation; note that the condition $x > 7$ from part (a) is exactly what makes every argument positive, so it is the domain, not decoration.

November 2025 TZ1 Paper 1 Q2 — 6 marks, no calculator.

```

q4a = ...          #  $x^2 - 8x + 7$  as a product
of two linear factors
q4b = [...]        # all roots of the equation
in (b)

verify_factored('4a', q4a,  $x^2 - 8x + 7$ ,
n=2)
verify_root_set('4b', q4b,
                  Eq(log( $x(x^2 - 1)$ ), 2) - 1,
log( $(x^2 - 8x + 7)(x + 1)$ ), 2)),
                  domain=Interval.open(7, oo))

```

Part II — downwards: out of the exponent

Theory 4. The laws of exponents, and the relation you were handed

$$\begin{aligned}a^m a^n &= a^{m+n}, \\ \frac{a^m}{a^n} &= a^{m-n}, \\ (a^m)^n &= a^{mn}, \\ a^0 &= 1.\end{aligned}$$

Nothing here is new. What is new is *where* the archive uses them: not to simplify an answer, but to make a substitution possible.

Suppose a question hands you $w(t) = Ce^{2t}$ together with the relation $e^{2M} = \frac{5}{C}$ from an earlier part, and then asks for $w(M + h)$. The whole of the two marks is one splitting:

$$w(M + h) = Ce^{2(M+h)} = C \underbrace{e^{2M}}_{= 5/C} e^{2h} = 5e^{2h}.$$

Two things go wrong here and both are visible in the line above. The coefficient inside the exponent has to reach *both* terms of $M + h$, so it is $e^{2M}e^{2h}$ and not $e^{2M}e^h$. And a relation is used in the direction it was given: $e^{2M} = \frac{5}{C}$, so the C cancels — writing e^{-2M} where e^{2M} belongs turns the fraction over and the answer with it.

"Deduce a similar expression" means do not start again. If the next part asks for $w(M - h)$, replacing h by $-h$ in the result is the whole of it: $5e^{-2h}$. Two marks, ten seconds, and the word *deduce* is the instruction.

Any positive number is a power of e . $3 = e^{\ln 3}$ — not a trick, just the definition of $\ln$ read backwards. May 2025 TZ1 Q7 opens with that one-mark part and then uses it to make sense of 3^{i-1} .

Task 5 ● — the substitution that does the work

The velocity of a particle is $v(t) = (1 + v_0)e^{-t} - 1$, and the time T at which its displacement is greatest satisfies $e^T = 1 + v_0$.

(a) Write down e^{-T} in terms of v_0 .

(b) Hence show that $v(T - k) = e^k - 1$, and **deduce** an expression for $v(T + k)$ in terms of k .

Part (a) is the step the markscheme pays for; the cell checks it, and then checks the deduction in (b).

May 2021 TZ2 Paper 1 Q11(c) and (d) — 4 marks, no calculator.

```
q5a = ...          # e^(-T) in terms of v0
q5b = ...          # v(T + k), in terms of k

verify_identity('5a', q5a, 1/(1 + v0),
var=v0)
verify_exact('5b', q5b, exp(-k) - 1)
```

Theory 5. Taking logarithms of both sides

When the unknown is in the exponent, no amount of algebra will get it out. One operation will, and it is the only one:

$$\begin{aligned}a^u &= b \\ \implies \\ u \ln a &= \ln b \\ \implies \\ u &= \frac{\ln b}{\ln a}.\end{aligned}$$

Isolate the power first. From $450 = 150e^{12k}$, divide by 150 *before* taking logarithms. $\ln(150e^{12k})$ is $\ln 150 + 12k$, not $150 \times 12k$, and half the lost marks in this technique are that line.

Two ways to say the same doubling time, and the markscheme prints both.

$$e^{-cT} = \frac{1}{2} \Rightarrow$$

$$-cT = \ln \frac{1}{2} =$$

$$-\ln 2$$

or

$$e^{cT} = 2 \Rightarrow cT = \ln 2.$$

The second avoids the minus sign entirely, which is why it is worth preferring: $\ln \frac{1}{2} = -\ln 2$ is the single most common sign slip in the topic.

Round at the end, never in the middle. When a question asks for a time to the nearest ten years, carrying a three-figure decimal for the rate instead of its exact logarithmic form moves the answer by years. Keep the exact form all the way to the last line — that is what "exact form" is *for*, even on a calculator paper.

"First exceeds" is a ceiling, not a rounding. If the inequality gives $n > 7.2$, the answer is 8 periods — and it is still 8 when the inequality gives $n > 7.8$. Rounding to the nearest is right half the

time and wrong the other half, which is the worst kind of habit.

Task 6 — a half-life, exactly and then numerically

The amount A of carbon-14 in a plant t years after its death is modelled by $A = A_0 e^{-kt}$, with $A_0, k > 0$. At the time of death a plant is defined to have 100 units. The half-life is 5730 years.

(a) Show that $k = \frac{\ln 2}{5730}$.

(b) Find, correct to the nearest 10 years, the time after death for 25% of the carbon-14 to decay.

(c) Write down the completed model $A(t)$, with both constants in place.

Part (c) is the first appearance of `verify_model`. It stores nothing: it puts $t = 0$, $t = 5730$ and $t = 11460$ back into whatever you wrote and asks whether 100, 50 and 25 come out. Only the numbers the question states go in — a prediction is checked by the question

that asks for it, not by the model's own definition.

May 2021 TZ2 Paper 2 Q5 — 7 marks, calculator allowed.

```
q6a = ...          # k, exactly
q6b = ...          # the time, to the nearest
10 years
q6c = ...          # A(t), an expression in t
with no letters left in it

verify_exact('6a', q6a, log(2)/5730)
check_num('6b', q6b, 3, 'b56267433ab0')
verify_model('6c', q6c, [(0, 100), (5730,
50), (11460, 25)])
```

Theory 6. Fitting a model: two constants, two conditions

$A(t) = A_0 e^{kt}$ has two unknowns, so it takes exactly two facts to pin down. The archive supplies them in the same shape every time.

1. **Decide what $t = 0$ is and write it down.** Not later — now. If the model starts in one year, the second fact is about another, and the question asks

about a third, then two subtractions decide everything that follows. Answering with the gap between the *second* and *third* dates is answering a different question perfectly.

2. **The first condition gives the multiplier.** Almost always it is $A(0)$, and $e^0 = 1$ makes it immediate: whatever the quantity was at the start **is** A_0 , with no algebra at all.
3. **The second gives k ,** by Theory 5. Here the phrasing does the damage: "*decreased by 30%*" means the new value is 0.70 of the old one, not 0.30 of it. Reading that line twice is worth more than any of the algebra after it.
4. **Substitute the target t .** Keep k exact until this line.

What the markscheme actually pays for. A seven-mark question of this shape splits as: recognising A_0 , turning the percentage into a multiplier, recognising the first t , substituting, finding k , substituting the target t , and the number.

Five of the seven are set-up. The exponential is one line in the middle.

How to check a fitted model in ten seconds. Put the data back in. If the model does not reproduce the very value it was fitted to, it is wrong regardless of how the algebra looked. That is exactly what `verify_model` does below, and it is a habit worth having on paper too.

Theory 7. Percentages, which are the same thing in English

A percentage change per period is a multiplier per period:

the question says	the multiplier is
grows by 8% a year	1.08 per year
depreciates at 12% per annum	0.88 per year
decreased by 30% over the period	0.70 over the period
nominal 6% per year, compounded monthly	$1 + \frac{0.06}{12} = 1.005$ per month
interest 2% a month against inflation 1.2% a month	$\frac{1.02}{1.012}$, or 1.0079... — see below

Nominal rate $\div$ number of periods. 6% compounded monthly is 0.5% twelve times, not 6% once. Questions of this shape make the division their own one-mark part precisely because everything after it depends on it.

The exponent counts periods, not years. After ten years of monthly compounding the exponent is 120, and this is where the archive puts its named wrong answer: see the solution to Task 7.

Inflation: the exam accepts both the exact version and the approximation. The *real* multiplier for an investment growing at i while prices grow at j is $\frac{1+i}{1+j}$. Markschemes print the subtraction $r = i - j$ as **METHOD 1** and the quotient as **METHOD 2**, and pay the same for either — for rates of this size the two differ around the fifth decimal place and land on the same period. Worth knowing which is which anyway: subtraction is a first-order approximation to the quotient,

and it stops being close when the rates get large.

On Paper 1 the power gets expanded, not evaluated. $400(1.05)^3$ is $400(1 + 3(0.05) + 3(0.05)^2 + (0.05)^3) = 400 + 60 + 3 + 0.05 = 463.05$, and a depreciation run **backwards** — dividing by the multiplier several times — is the Maclaurin series of $(1 - x)^{-m}$ at a small x . That is the binomial theorem from A3 doing service here, and it is the reason questions like these are set on Paper 1 at all.

Task 7 — a whole Paper 2 model question

Alex purchases a car for €30 000. The value of the car depreciates at 15% per annum.

(a) Find the value of the car after ten years, correct to two decimal places.

Alex invests €50 000 in a bank account paying compound interest of 1.5% per month. Inflation over the same period was 0.8% per month.

(b) Find the number of months required for the **real** value of the investment to first exceed €55 000.

May 2025 TZ2 Paper 2 Q4 — 6 marks, calculator allowed.

```
q7a = ...          # the value of the car after  
ten years  
q7b = ...          # the number of months  
  
check_num('7a', q7a, 6, '774db6f92305')  
check_num('7b', q7b, 2, '8527a891e224')
```

Task 8 — the same percentages with no calculator

Bob invests 1000 dinar in an account paying a nominal annual interest rate of 4% compounded **quarterly**. The amount after one complete year can be written as $1000(1 + k)^4$, $k \in \mathbb{Q}$.

(a) Write down the value of k .

(b) Hence or otherwise find the amount in the account after one complete year, to the nearest dinar.

A car depreciated in value by 10% each year and is worth \$1000 today. The first four terms of the Maclaurin expansion of $(1 - x)^{-4}$ are $1 + 4x + 10x^2 + 20x^3$.

(c) Estimate the value of the car four years ago.

May 2025 TZ1 Paper 1 Q2 — 5 of its 7 marks. November 2025 TZ1 Paper 1 Q5(b) — 2 marks. Both Paper 1.

```
q8a = ...          # k, exactly
q8b = ...          # the amount after one year,
to the nearest dinar
q8c = ...          # the value of the car four
years ago

verify_exact('8a', q8a, Rational(1, 100))
check_num('8b', q8b, 4, '8c6c42f379f0')
check_num('8c', q8c, 3, 'b3f2d86521d7')
```

Theory 8. The logistic model

$$x(t) = \frac{L}{1 + Ce^{-kt}}, \quad L, C, k > 0.$$

Three constants, and the archive gives you them in a fixed order.

L is read off the page. It is the value the quantity settles at as $t \rightarrow \infty$, because $e^{-kt} \rightarrow 0$. When a question says a population "*had a stable size of*" something and then tells you to take that as L , the assumption being made is that the long-run size will be what it was before — and stating that assumption is itself a mark.

Take a colony that levels off at $L = 5000$, starts at 500 and reaches 1500 after three hours.

C comes from $t = 0$, and it is easy because $e^0 = 1$:

$$\begin{aligned} 500 &= \frac{5000}{1 + C} \\ \implies 500 + 500C &= 5000 \\ \implies C &= 9. \end{aligned}$$

k comes from the second data point, by Theory 5 again: $1500 = \frac{5000}{1 + 9e^{-3k}}$ gives $e^{-3k} = \frac{7}{27}$, so $k = \frac{1}{3} \ln \frac{27}{7} = 0.450$.

Then predict, and round to whole units. $x(6) = 3115.38 \dots \rightarrow 3115$.

The ten-second sanity check. Evaluate your model at a large t . It must approach L from below. If it runs away or falls, the sign of k is wrong — and a wrong sign there is invisible in the algebra and obvious in the number.

Task 9 **— a whole Paper 3 fit**

A wolf population had a stable size of 200. After a disruption it fell to 40, the area became protected, and the population began to grow again. Let x be the size of the population t years after protection began, modelled by $x = \frac{L}{1 + Ce^{-kt}}$ with $L = 200$.

At $t = 0$ there are 40 wolves; at $t = 5$ there are 70.

(a) Find C . **(b)** Find k , to three significant figures.

(c) Write down the completed model.

(d) Predict the population at $t = 10$, to the nearest whole number.

November 2025 TZ1 Paper 3 Q2(a) — 7 of the question's 29 marks, calculator allowed.

```
q9a = ...          # C
q9b = ...          # k
q9c = ...          # x(t), an expression in t
                    # with no letters left in it
q9d = ...          # the population at t = 10

verify_exact('9a', q9a, 4)
check_num('9b', q9b, 3, 'e5bd4a550198')
verify_model('9c', q9c, [(0, 40), (5, 70)])
check_num('9d', q9d, 3, '3346f2bbf6c3')
```

Part III — the logarithm as a function

Theory 9. Composition, domain, and scale

The first eight techniques treat $\ln$ as a tool. This one treats it as a function with a graph, a domain and a range, and the archive asks about all three.

Composition. $(f \circ g)(x) = f(g(x))$: put g inside f . With $f(x) = x^2 - 4$ and $g(x) = \ln x$,

$$(f \circ g)(x) = (\ln x)^2 - 4.$$

Solving an inequality on such a composite is a two-stage job, and doing it in one stage is how the mark is lost.

Solve for $\ln x$ first: from $(\ln x)^2 - 4 \leq 0$ comes $-2 \leq \ln x \leq 2$, and only then exponentiate, which is allowed because e^u is increasing and so preserves the inequality:

$$e^{-2} \leq x \leq e^2.$$

Applying e to -2 and getting $-e^2$ is the standard error, and it is the same error as writing $\ln \frac{1}{2} = -\ln 2$ backwards.

Domain of a composition. $\text{dom}(g \circ f)$ is the set of x in the domain of f whose image $f(x)$ lies in the domain of g . With $g = \ln$ that means $f(x) > 0$, so for the same f ,

$$x^2 - 4 > 0 \iff x < -2 \text{ or } x > 2.$$

Note what it is *not*: the domain of f , which is all of $\mathbb{R}$. The composition is narrower than either function alone, and the marks are split between finding the critical values and getting the two inequalities the right way round.

A logarithmic scale. A scale of the form $S = c \log_{10}(\text{something} \times 10^m)$ turns a huge range of values into a small one, and two facts follow immediately from Theory 1 without a calculator:

- multiplying the quantity inside by k adds $c \log_{10} k$ to the reading — it does not multiply the reading by k ;

- adding c to the reading multiplies the quantity by 10.

That is what a log scale is *for*: it turns multiplication into addition. Every question about one is Theory 1 wearing a lab coat.

And an equation given as $\ln(g(x)) = \dots$ is not an equation for x — it is a definition of g , one exponential away:

$$\ln(g(x)) = 2x \ln 3 = \ln(3^{2x}) \implies g(x) = 9^x.$$

The power law does the work — a coefficient in front of a logarithm goes up into the exponent — and exponentiating is the last step, not the first.

Task 10 — **composition, both ways**

The function f is defined by $f(x) = 5(x + 1)(x + 3)$, $x \in \mathbb{R}$; the function g by $g(x) = \ln x$, $x > 0$. You may use that $f(x) \leq 40 \iff -5 \leq x \leq 1$.

(a) Write down an expression for $(f \circ g)(x)$.

(b) Solve $(f \circ g)(x) \leq 40$.

(c) Find the domain of $g \circ f$.

May 2025 TZ1 Paper 1 Q10(d) and (e) — 6 marks, no calculator.

```
q10a = ...      # (f o g)(x)
q10b = ...      # the solution set of (f o
g)(x) <= 40
q10c = ...      # the domain of g o f

verify_identity('10a', q10a, 5*(log(x) + 1)*
(log(x) + 3))
verify_solution_set('10b', q10b, 5*(log(x) +
1)*(log(x) + 3) <= 40,
                    domain=Interval.open(0,
oo))
verify_solution_set('10c', q10c, 5*(x + 1)*(x
+ 3) > 0)
```

Task 11 — a scale where multiplying is adding

The loudness L of a sound, in decibels, is related to its intensity I units by $L = 10 \log_{10}(I \times 10^{12})$. Sound S_1 has intensity 10^{-6} units and loudness 60 decibels. Sound S_2 has twice the intensity of S_1 .

(a) State the intensity of S_2 .

(b) Determine the loudness of S_2 .

(c) The maximum loudness of thunder was measured at 115 decibels. Find the corresponding intensity.

May 2024 TZ2 Paper 2 Q3 — 6 marks, calculator allowed. Part (a) is exact; (b) and (c) to three significant figures.

```
q11a = ...      # the intensity of S2,
exactly
q11b = ...      # the loudness of S2
q11c = ...      # the intensity of the
thunder

verify_exact('11a', q11a, 2*Rational(1,
10**6))
check_num('11b', q11b, 3, 'da4ea2a5506f')
check_num('11c', q11c, 3, '6d7847027d4f')
```

Task 12 — an exponential hiding behind a logarithm

Let $M_n(S)$ be the largest possible product of n positive reals with sum S . It has been proved that $M_n(S) = \left(\frac{S}{n}\right)^n$.

(a) Write down $M_3(12)$, $M_4(12)$ and $M_5(12)$, exactly.

The function g is defined by $\ln(g(x)) = x \ln \frac{S}{x}$, for $x \in \mathbb{R}^+$.

(b) Find $g(x)$, and so verify that $g(x) = M_x(S)$ for $x \in \mathbb{Z}^+$.

The point of (b) is that the formula for a *whole number* of factors extends to a real x — which is what makes it differentiable, and is why the rest of that Paper 3 question can maximise it.

May 2023 TZ1 Paper 3 Q2(e) and (i) — 5 marks, calculator allowed, but every answer here is exact.

```
q12a3 = ...      # M_3(12), exactly
q12a4 = ...      # M_4(12), exactly
q12a5 = ...      # M_5(12), exactly
q12b = ...       # g(x), in terms of S and x

verify_exact('12a-3', q12a3, 4**3)
verify_exact('12a-4', q12a4, 3**4)
verify_exact('12a-5', q12a5, Rational(12,
5)**5)
verify_identity('12b', q12b, exp(x*log(S/x)),
var=x)
```

The same question on Paper 2

May 2021 TZ2 Paper 2 Q5(c) — *find, correct to the nearest 10 years, the time for 25% of the carbon-14 to decay* — is the one place in this topic where the markscheme prints both routes side by side and pays the same for either:

EITHER using an appropriate graph to attempt to solve for t **(M1)** **OR** manipulating logs to attempt to solve for t **(M1)** $\ln 0.75 = -\frac{\ln 2}{5730}t$

So the calculator is not a shortcut past the mathematics; it is one of two accepted methods, and the marks for setting up $75 = 100e^{-kt}$ come first either way. The cell below does both and prints them together.

The interesting part is what the calculator does **not** give you: the exact form. Notice that the graph route lands on a decimal and stops, while the algebra route lands on an expression built out of logarithms, which you can then round to

whatever the question asks for. On Paper 2 that difference costs nothing. In part (b) of the same question, where the answer is $\frac{\ln 2}{5730}$, it costs the whole mark.

```
kc = log(2)/5730
model = 100*exp(-kc*t)

exact = list(solveset(Eq(model, 75), t,
Interval(0, oo)))[0]
print('exact time for 25% to decay:', exact,
'      ( = 5730*ln(4/3)/ln 2 )')
print('as a decimal:                ',
nsolve(Eq(model, 75), t, 2000))
print('to the nearest 10 years:      ',
10*round(float(nsolve(Eq(model, 75), t,
2000))/10))
print()
print('and the check that the model is the
right one:')
verify_model('by graph', model, [(0, 100),
(5730, 50)])
```

Trainer: name the technique in five seconds

Twelve openings. For each one, name the technique — **do not solve anything**. This is the cheapest half-hour in the whole practicum: on the exam the

difference between a solved question and an unsolved one is usually the first thirty seconds.

Use these codes:

`laws` · the three log laws | `base` · change of base | `logeq` · log equation | `powers` · laws of exponents | `takelogs` · take logs of both sides | `fit` · fit a model to data | `percent` · percentages as a power | `logistic` · the logistic model | `function` · the logarithm as a function

1. Let $\log_{10} 2 = p$. Find $\log_{10} 50$ in terms of p .
2. Solve $\log_9 x + \log_3 x = 6$.
3. Solve $\log_5(x + 3) + \log_5(x - 1) = 1$.
4. Given $e^{2a} = 7$, write e^{-6a} as a fraction.
5. Solve $3^{x+1} = 200$, giving your answer to three significant figures.
6. A culture has 500 bacteria at noon and 1200 at 3 pm and grows as $N = N_0 e^{kt}$. Estimate the count at 6 pm.
7. A machine bought for \$8000 loses 12% of its value each year. Find its value

after seven years.

8. A fish population cannot exceed 5000; it was 400 when counting began and 900 after two years. Predict year five.
9. Find the domain of $h(x) = \ln(x^2 - 4)$.
0. The half-life of a substance is 12 hours. Find the time for 90% of it to decay.
1. Solve $\log_4 x = \log_{16}(x + 6)$.
2. A town of 20 000 shrinks by 3% per year for eight years, then is modelled as Pe^{kt} from that point. Find k .

```
answers = {  
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',  
    7: '', 8: '', 9: '', 10: '', 11: '', 12:  
    '',  
}  
  
trigger_check(answers, {1: '8954b4ce200e', 2:  
'cae662172fd4', 3: 'eedf039eb95b', 4:  
'0e25d57dbe55', 5: '5aca8050e55a', 6:  
'da4270e3735a', 7: 'fd86717aca41', 8:  
'03a311656c69', 9: '78f9ac018e55', 10:  
'5aca8050e55a', 11: 'cae662172fd4', 12:  
'da4270e3735a'})
```

On the timer

November 2022 Paper 2 Q4 — 7 marks. Target: 11 minutes.

Close everything above. Calculator allowed.

The population of a town t years after 1 January 2014 can be modelled by

$$P(t) = 15\,000e^{kt}, \quad k < 0, t \geq 0.$$

It is known that between 1 January 2014 and 1 January 2022 the population decreased by 11%.

Use this model to estimate the population of this town on 1 January 2041.

Seven marks for one sentence, and five of them are set-up. Before touching the calculator, write down three things: what $t = 0$ is, what t is on 1 January 2022, and what t is on 1 January 2041.

Attempt log. Copy this line and add a date and a time each pass.

2026-09-01 —

```
qt_model = ...    # the population model, with  
k in place  
qt = ...          # the population on 1  
January 2041  
  
verify_model('timed model', qt_model, [(0,  
15000), (8, 13350)])  
check_num('timed answer', qt, 3,  
'a91bb59c3f9c')
```

Solutions

Read one only after you have an answer of your own — even a wrong one. Each solution names its source and the marks it carries.

Task 1 — the laws in both directions

May 2025 TZ3 P1 Q1(a), 3 marks · May 2025 TZ1 P1 Q8(a), 2 marks

(a) Factorise first, because there is no law for a sum:

$$\begin{aligned}24 &= 2^3 \times 3 \\&\implies \log_{10} 24 \\&= \log_{10} 2^3 + \log_{10} 3 \\&= 3 \log_{10} 2 + \log_{10} 3 \\&= 3p + q.\end{aligned}$$

The markscheme awards **(M1)** for using $\log ab = \log a + \log b$ or $\log a^m = m \log a$, **(A1)** for reaching $3 \log_{10} 2 + \log_{10} 3$ with the arguments actually being 2 and 3, and **A1** for $3p + q$.

(b) The loose 1 has to join the logarithm:

$$1 + \log_2 n = \log_2 2 + \log_2 n = \log_2(2n).$$

Then, since $n \geq 1$ gives $2n \geq n + 1$, and since $\log_2$ is **increasing**,

$$\log_2(2n) \geq \log_2(n + 1),$$

which is the statement. The second half is an **R1** and it is for naming monotonicity out loud; the inequality $2n \geq n + 1$ alone does not earn it.

Task 2 — one base at a time

May 2025 TZ3 P1 Q1(b), 2 marks · May 2024 TZ1 P1 Q2, 5 marks

(a)
$$\log_3 8 = \frac{\log_{10} 8}{\log_{10} 3} = \frac{3 \log_{10} 2}{\log_{10} 3} = \frac{3p}{q}.$$

Argument on top, base underneath.

(b)
$$\log_{10} \frac{1}{a} = \log_{10} 1 - \log_{10} a = -\log_{10} a = -\frac{1}{3}.$$

Equivalently $\log_{10} a^{-1} = -\log_{10} a.$

(c)
$$\log_{1000} a = \frac{\log_{10} a}{\log_{10} 1000} = \frac{1/3}{3} = \frac{1}{9}.$$
 The three in the denominator is $\log_{10} 10^3$ — the same shortcut as $\log_{b^m} a = \frac{1}{m} \log_b a.$

Task 3 — two bases, one unknown

November 2025 TZ3 P1 Q4, 5 marks

Everything is a power of 2: $8 = 2^3$ and $4 = 2^2$. So

$$\begin{aligned}3 \log_8 10x &= 3 \cdot \frac{\log_2 10x}{3} \\&= \log_2 10x, & \log_4 x \\&= \frac{\log_2 x}{2} \\&= \log_2 x^{1/2} \\&= \log_2 \sqrt{x}.\end{aligned}$$

The equation becomes

$$\begin{aligned}\log_2 10x - \log_2 \sqrt{x} &= 1 \\&\implies \log_2 \frac{10x}{\sqrt{x}} \\&= \log_2 2 \\&\implies 10\sqrt{x} \\&= 2,\end{aligned}$$

$$\text{so } \sqrt{x} = \frac{1}{5} \text{ and } x = \frac{1}{25}.$$

Both **M** marks are for method seen anywhere — power/quotient rule and change of base — and they are

independent, so partial credit survives a wrong order. The last three marks are the correct equation in one base, the correct equation without logarithms, and the answer. Note the trap that the mark scheme implies by printing $10\sqrt{x} = 2$ separately: stopping at $\sqrt{x} = \frac{1}{5}$ answers a different question.

Task 4 — the loose -1 , and the domain

November 2025 TZ1 P1 Q2, 6 marks

(a) $x^2 - 8x + 7 = (x - 7)(x - 1)$ and $x^2 - 1 = (x - 1)(x + 1)$, so

$$\frac{x}{(x - 7)(x - 1)} \times \frac{(x - 1)(x + 1)}{x + 1} = \frac{x}{x - 7}.$$

(b) The -1 is $\log_2 \frac{1}{2}$, or equivalently move it across as $1 = \log_2 2$. Collapsing both sides,

$$\log_2 \frac{x(x^2 - 1)}{(x^2 - 8x + 7)(x + 1)} = \log_2 2.$$

The fraction inside is exactly what part (a) simplified, so it is $\frac{x}{x-7}$, and removing the logarithms gives

$$\frac{x}{x-7} = 2 \implies x = 2x - 14 \implies x = 14.$$

$14 > 7$, so every argument is positive and the root stands.

Why the domain matters even though nothing was discarded here. The condition $x > 7$ came from part (a) and is what makes $x^2 - 8x + 7 > 0$. Had the algebra produced a second root below 7, it would have been manufactured by the removal step, not present in the original equation — and `verify_root_set` says so in exactly those words.

Task 5 — the substitution that does the work

May 2021 TZ2 P1 Q11(c) and (d), 4 marks

(a) $e^T = 1 + v_0$, so $e^{-T} = \frac{1}{1 + v_0}$.

Reciprocal, not the same thing — this is

the one line the two marks of (c) are built on.

(b) Split the exponent, then substitute:

$$\begin{aligned}v(T - k) &= (1 + v_0)e^{-(T-k)} - 1 \\&= (1 + v_0)e^{-T}e^k - 1 \\&= (1 + v_0) \cdot \frac{1}{1 + v_0} \cdot e^k - 1 \\&= e^k - 1.\end{aligned}$$

The markscheme's METHOD 2 is shorter still: write $1 + v_0$ as e^T from the start, and $e^T e^{-(T-k)} = e^k$ falls out in one step.

For (d), *deduce*: replacing k by $-k$ turns " k seconds before" into " k seconds after", so $v(T + k) = e^{-k} - 1$. Two marks for one substitution.

And the part (e) that follows, though it is outside this practicum: $v(T - k) + v(T + k) = e^k + e^{-k} - 2 \geq 0$ because $e^k + e^{-k} \geq 2$ by AM-GM. The particle is faster before the turning point than it is slow after it.

Task 6 — a half-life, exactly and then numerically

May 2021 TZ2 P2 Q5, 7 marks

(a) Half the original amount after 5730 years:

$$\begin{aligned}50 &= 100e^{-5730k} \\&\implies e^{-5730k} \\&= \frac{1}{2} \\&\implies e^{5730k} \\&= 2 \\&\implies 5730k \\&= \ln 2 \\&\implies k \\&= \frac{\ln 2}{5730}.\end{aligned}$$

The markscheme's note is unusually generous — *"award full marks for at least two correct algebraic steps seen"* — and the reason is that there are several routes. The route above avoids $\ln \frac{1}{2}$ entirely.

(b) 25% has decayed means 75 units **remain**, not 25:

$$\begin{aligned}
75 &= 100e^{-\frac{\ln 2}{5730}t} \\
&\implies \ln 0.75 \\
&= -\frac{\ln 2}{5730}t \\
&\implies t \\
&= \frac{5730 \ln \frac{4}{3}}{\ln 2} \\
&= 2378.16 \dots
\end{aligned}$$

To the nearest 10 years, $t = 2380$.

(c) $A(t) = 100e^{-\frac{\ln 2}{5730}t}$, equivalently $100 \cdot 2^{-t/5730}$. Both pass the check, because the check puts $t = 0$, $t = 5730$ and $t = 11460$ back in and asks for 100, 50 and 25.

Task 7 — a whole Paper 2 model question

May 2025 TZ2 P2 Q4, 6 marks

(a) Depreciation at 15% is a multiplier of 0.85:

$$30\,000 \times 0.85^{10} = 5906.2321 \dots \rightarrow \text{€}5906.23.$$

Two decimal places, because the question says so and because it is money.

(b) The real multiplier per month, exactly, is a **quotient**:

$$\frac{1 + 0.015}{1 + 0.008} = \frac{1.015}{1.008} = 1.00694 \dots$$

Then

50

$$000(1.00694 \dots)^n > 55$$

$$000 \implies n > \frac{\ln 1.1}{\ln 1.00694 \dots} = 13.7722 \dots$$

so the first month is $n = 14$.

The markscheme's own **METHOD 1** is the approximation $r = 1.5 - 0.8 = 0.7\%$, which gives $n > \frac{\ln 1.1}{\ln 1.007} = 13.6633 \dots$ and the same answer, and it earns full marks. Either is fine. It also awards a mark for one correct crossover value: with 0.7% , $n = 13$ gives €54 746.09 and $n = 14$ gives €55 129.31.

The wrong answer the markscheme names. 164 months, worth **(A1)(M1)** **(A0)(A0)** — the method is right and both answer marks are gone. It comes from reading 0.7% as an *annual* rate and

compounding it monthly: $\frac{\ln 1.1}{\ln(1+0.007/12)} = 163.5\dots$ The period was in the question all along; the arithmetic never noticed.

Task 8 — the same percentages with no calculator

May 2025 TZ1 P1 Q2, 5 of 7 marks

November 2025 TZ1 P1 Q5(b), 2 marks

(a) *Nominal 4% compounded quarterly* means the rate per quarter is $\frac{4\%}{4} = 1\%$, so $k = \frac{4}{400} = \frac{1}{100} = 0.01$. One mark, and everything downstream fails without it.

(b) No calculator, so expand rather than evaluate. With $(1+x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$ at $x = \frac{1}{100}$,

$$\begin{aligned} 1000 \left(1 + \frac{1}{100}\right)^4 &= 1000 \\ &+ 40 \\ &+ 0.6 \\ &+ 0.004 \\ &+ \dots \\ &= 1040.6\dots \rightarrow 1041 \text{ dinar.} \end{aligned}$$

The first two terms already give 1040; the third decides the rounding.

(c) Four years of 10% depreciation multiply by 0.9^4 , so the value four years ago is $1000 \times 0.9^{-4} = 1000(1 - 0.1)^{-4}$. With the given expansion at $x = 0.1$,

$$\begin{aligned}(1 - x)^{-4} &\approx 1 \\ &+ 4(0.1) \\ &+ 10(0.01) \\ &+ 20(0.001) \\ &= 1.52 \\ &\implies \$1520.\end{aligned}$$

(The exact value is $1000/0.6561 = 1524.16\dots$; the series truncated at x^3 gives 1520, which is what the question asks for.)

Task 9 — a whole Paper 3 fit

November 2025 TZ1 P3 Q2(a), 7 marks

(a) At $t = 0$, $e^0 = 1$:

$$\begin{aligned}
40 &= \frac{200}{1+C} \\
&\implies 40 + 40C \\
&= 200 \\
&\implies C \\
&= 4.
\end{aligned}$$

(b) At $t = 5$:

$$\begin{aligned}
70 &= \frac{200}{1+4e^{-5k}} \\
&\implies 1+4e^{-5k} \\
&= \frac{20}{7} \\
&\implies e^{-5k} \\
&= \frac{13}{28} \\
&\implies k \\
&= \frac{1}{5} \ln \frac{28}{13} \\
&= 0.153451\dots \rightarrow 0.153.
\end{aligned}$$

The markscheme prints it as $-\frac{1}{5} \ln \frac{130}{280}$, which is the same number written with the minus still inside.

$$\textbf{(c)} \quad x(t) = \frac{200}{1+4e^{-0.153451\dots t}}.$$

(d) $x(10) = \frac{200}{1 + 4e^{-1.53451\dots}} = 107.397\dots \rightarrow$
 107 wolves. Whole animals, so a whole number.

What the rest of that question does with this, for context: it compares the logistic prediction against a Gompertz model, which satisfies $\frac{dx}{dt} = ax \ln \frac{200}{x}$ and integrates — by separation of variables, technique 2 of E7 — to $\ln x = \ln 200 - Ae^{-at}$. That model predicts 101. The measured population turned out to be 85, so **both models overestimate and the Gompertz one is closer**, which is the one-mark part (c).

Task 10 — composition, both ways

May 2025 TZ1 P1 Q10(d) and (e), 6 marks

(a) $(f \circ g)(x) = f(\ln x) = 5(\ln x + 1)(\ln x + 3)$.
 The markscheme also accepts $5(\ln x + 2)^2 - 5$ and $5(\ln x)^2 + 20 \ln x + 15$.

(b) Part (c) of the same question established $f(x) \leq 40 \iff -5 \leq x \leq 1$.

Replacing x by $\ln x$,

$$-5 \leq \ln x \leq 1 \implies e^{-5} \leq x \leq e.$$

Exponentiating preserves the inequalities because e^u is increasing. The **(M1)** is precisely for "*attempt to replace x with $\ln x$ using their solution to part (c)*" — that is, for not solving the quadratic again.

(c) $(g \circ f)(x) = \ln(f(x))$ needs $f(x) > 0$:

$$(x + 1)(x + 3) > 0$$

$$\implies x <$$

$$-3 \text{ or } x >$$

$$-1,$$

that is $(-\infty, -3) \cup (-1, \infty)$. Two separate marks: one for the critical values -3 and -1 , one for the inequalities pointing the right way. Writing $-3 < x < -1$ — the *inside* of the parabola, where f is negative — collects the first and loses the second.

Task 11 — a scale where multiplying is adding

May 2024 TZ2 P2 Q3, 6 marks

(a) Twice 10^{-6} is 2×10^{-6} . One mark, and it is a mark for reading.

(b)

$$\begin{aligned} L &= 10 \log_{10}(2 \times 10^{-6} \times 10^{12}) \\ &= 10 \log_{10}(2 \times 10^6) \\ &= 10 (6 + \log_{10} 2) \\ &= 60 + 10 \log_{10} 2 \\ &= 63.0102 \dots \rightarrow 63.0. \end{aligned}$$

The markscheme accepts $60 + 10 \log_{10} 2$ as a final answer. Note what did **not** happen: doubling the intensity did not double the loudness, it added about 3 decibels. That is the whole idea of a log scale.

(c)

$$\begin{aligned}
115 &= 10 \log_{10}(I \times 10^{12}) \\
&\implies I \times 10^{12} \\
&= 10^{11.5} \\
&\implies I \\
&= 10^{-0.5} \\
&= \frac{1}{\sqrt{10}} \\
&= 0.316227 \dots \rightarrow 0.316.
\end{aligned}$$

Exact forms $10^{-0.5}$ and $\frac{1}{\sqrt{10}}$ are accepted too.

Task 12 — an exponential hiding behind a logarithm

May 2023 TZ1 P3 Q2(e) and (i), 5 marks

(a) Straight into $\left(\frac{S}{n}\right)^n$ with $S = 12$:

$$M_3(12) = 4^3 = 64,$$

$$M_4(12) = 3^4 = 81,$$

$$M_5(12) = 2.4^5 = 79.62624.$$

Three separate one-mark parts, and the sequence is the point: the product rises, peaks somewhere between $n = 4$ and $n = 5$, and falls. The next parts of that

question find the peak — it is at $x = S/e$, which for $S = 12$ is 4.41.

(b) Use the power law on the right-hand side and then exponentiate:

$$\begin{aligned}\ln(g(x)) &= x \ln \frac{S}{x} \\ &= \ln \left(\frac{S}{x} \right)^x \\ \implies g(x) &= \left(\frac{S}{x} \right)^x,\end{aligned}$$

which is $M_x(S)$ whenever x is a positive integer. **M1** for moving the x up as a power, **A1** for g itself.

Why anyone would bother. $M_n(S)$ is defined only for whole n — you cannot have 4.41 factors. Writing it as g , a function of a real variable that agrees with M_n at the integers, is what makes it differentiable, and differentiating is how the question then finds the best n . Extending a discrete formula to a continuous one by way of logarithms is a move worth recognising; it is the same

move that turns compound interest into e^{rt} .

On the timer

November 2022 Paper 2 Q4, 7 marks

$t = 0$ is 1 January 2014, so 1 January 2022 is $t = 8$ and 1 January 2041 is $t = 27$.

A decrease of 11% leaves 89%:

$$P(8) = 0.89 \times 15$$

$$000 = 13$$

$$350 \implies 15$$

$$000e^{8k} = 13$$

$$350 \implies e^{8k} = 0.89 \implies k = \frac{\ln 0.89}{8} = -0.014566 \dots$$

Then

$$P(27) = 15\,000e^{-0.014566 \dots \times 27} = 10\,122.3 \dots$$

The markscheme accepts 10 100 (three significant figures) or 10 122.

Where the seven marks are.

Recognising $P(0) = 15\,000$; computing $0.89 \times 15\,000$; recognising $t = 8$; substituting both into the model; $k = \frac{\ln 0.89}{8}$; substituting $t = 27$; the number. Six

of the seven are before the exponential is ever evaluated, and the two most commonly lost are the third and the sixth — the two that are only about what t means.

What to do when it goes wrong

Your model gives the wrong number at a data point. The constants are not the problem — the time is. Write down what $t = 0$ is and recount.

Your answer is off by a factor you cannot find. Check whether a percentage went in as 0.11 where 0.89 was meant, or as 1.04 where 1.01 was meant.

Your logarithmic equation has a root the check rejects. Removing logarithms is not reversible. Put the root back into the *original* equation and look at every argument.

Your inequality came out backwards. You applied e^u or $\ln$ to one side only, or to

a negative number. Both are increasing functions: they never flip an inequality, and they never accept a negative argument.

Your exact answer became a decimal.

On Paper 1 that is the whole mark. Keep $\ln 2$, keep $\frac{1}{9}$, keep $3p + q$; the calculator comes out only when the paper number says it may.